

SUMMARY

Machine Learning Engineer with 3+ years of experience building production recommender systems, ranking models, distributed ML pipelines, and experimentation platforms at Dream11. Strong background in PySpark, Databricks, MLflow, Kafka, Gaussian-process bandits, sequential recommenders, A/B testing, and scientific machine learning.

EDUCATION

University of Wisconsin-Madison

M.S. in Physics, GPA: 4.00/4.00

Indian Institute of Technology Madras

B.Tech. in Electrical Engineering

Madison, WI

Aug 2024 - Dec 2025

Chennai, India

Jul 2016 - Jun 2020

EXPERIENCE

Dream11

Senior Machine Learning Engineer

Mumbai, India

Jul 2020 - Sep 2023

- Built end-to-end personalization and ranking systems for fantasy contests and sports-match recommendations using Databricks, Redshift, Delta Lake, and Kafka; served 10M+ daily users and drove about 3% revenue lift in A/B tests.
- Designed distributed ML pipelines using PySpark, Horovod, SynapseML, AWS, Spark Streaming, and MLflow for model training, batch and near-real-time inference, versioning, and experimentation.
- Built real-time monitoring and experimentation dashboards with Slack alerting using Spark Streaming and Kibana to track model performance, recommendation quality, A/B-test lifts, and production anomalies.
- Implemented and evaluated ranking and recommendation models including list-wise ranking, sequential recommenders, matrix factorization, and graph-based models for personalized user representation learning.
- Formulated continuous recommender-system tuning as a Gaussian-process bandit problem, using noisy online experiment feedback as the reward signal to optimize production parameters.
- Led and analyzed 20+ production A/B tests across models, UI, pricing, and feature sensitivity in collaboration with product, data, and engineering teams.

University of Wisconsin-Madison

Volunteer Research Assistant, Computational Cosmology / Scientific ML

Madison, WI

May 2026 - Present

- Exploring CosmoFlow-style scientific-computing methods for inflationary cosmology, with focus on Hamiltonian-flow formulations for correlators and Python-based numerical workflows.
- Studying perturbation theory, coupled dynamical systems, and ODE-based approaches for theoretical-physics applications in QFT and cosmology.

University of Wisconsin-Madison

Research Assistant, Computational Holography

Madison, WI

Jan 2025 - Sep 2025

- Implemented PyTorch-based iterative and gradient-based optimization pipelines for phase retrieval and hologram generation in a neutral-atom quantum computing experiment.
- Improved optical trap intensity, waist, and position uniformity to about 99.5% through optical apodization, Gaussian fitting, and quantitative benchmarking.
- Explored Gaussian-process Bayesian optimization over Zernike coefficients for data-driven optical aberration correction.

Samsung R&D Institute

Research Intern

Noida, India

May 2019 - Jul 2019

- Built an end-to-end ultrasonic hand-gesture detection pipeline on Android using signal processing, CNN models, and TensorFlow Lite deployment.

TECHNICAL SKILLS

Machine Learning	Recommender systems, learning-to-rank, list-wise ranking, sequential recommenders, matrix factorization, graph-based models, representation learning, bandits, Bayesian optimization, A/B testing
Data & Platforms	PySpark, Databricks, Delta Lake, Redshift, MLflow, Kafka, Spark Streaming, AWS, SageMaker, Kibana, Docker, Git, Linux
Programming	Python, SQL, C/C++, Julia, PyTorch, TensorFlow, SynapseML, Horovod, scikit-learn, pandas, NumPy, SciPy, numerical optimization
Coursework	Quantum Computing, Quantum Algorithms, Quantum Mechanics I & II, Electrodynamics, General Relativity, Quantum Field Theory I

ACHIEVEMENTS

Indian National Physics Olympiad Gold Medal | JEE Advanced All India Rank 586 | KVPY Fellowship | Phi Kappa Phi Honor Society